

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000797		
	Version:	4.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	08-oct-2012
Title:	SM DG57-DG517. Samples-Reagents carousel module verification.			

DG-57 / DG517 Service Manual

SAMPLES-REAGENT CARROUSEL MODULE VERIFICATION

Procedure description
This procedure describes how to check the Samples-Reagent carousel module parts: home detectors, positioning performance, reagents stirring, module settings and correct reagents spinning. <i>Note: Document REGD-0000797_v4 replaces document REGD-0000797_v3.</i>

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EQUIPMENT

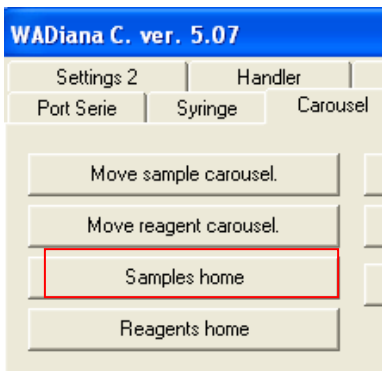
1. Digital photo tachometer.
2. Reflective tape.

SET UP

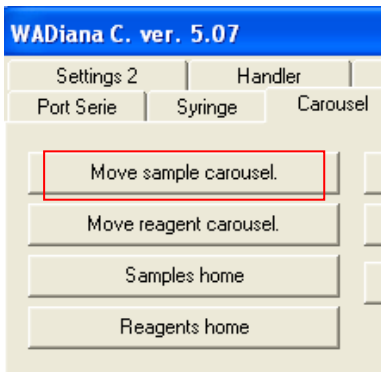
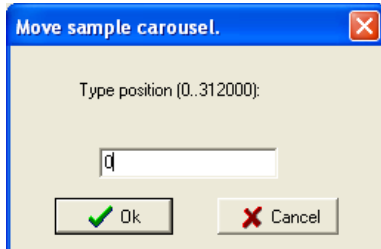
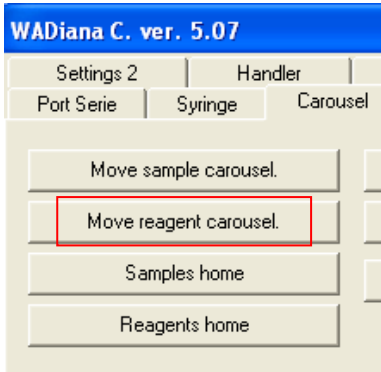
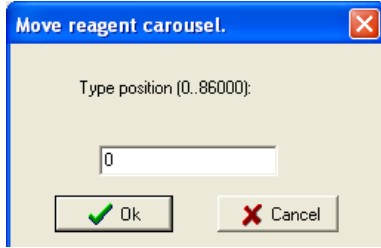
1. Turn on the instrument.
2. Execute the ST software (Technical Service software).
3. Enter the Technical Service password (d+g_diana).

PROCEDURE 1 - Checking homes

Note: this procedure can be executed with **No Init Technical Service software** also.

Process	Expected Results
1. Access 'Carousel' tab in ST software. 2. Press the 'Samples home' button.  Picture 1	--- The samples carousel spins counterclockwise up to 1 turn. It can move slightly clockwise followed by a slightly counterclockwise movement if the carousel was at home position already.
3. Press the 'Reagents home' button.  Picture 2	The reagents carousel spins counterclockwise up to 1 turn. It can move slightly clockwise followed by a slightly counterclockwise movement if the carousel was at home position already.

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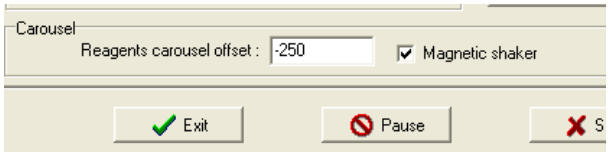
Process	Expected Results
4. Press the 'Move sample carousel' button.  Picture 3 5. Type '0' in the text box and press the 'Ok' button.  Picture 4 6. Press the 'Move reagent carousel' button.  Picture 5 7. Type '0' in the edit box and press the 'Ok' button.  Picture 6	--- The samples carousel moves slightly. The reagents carousel moves slightly.

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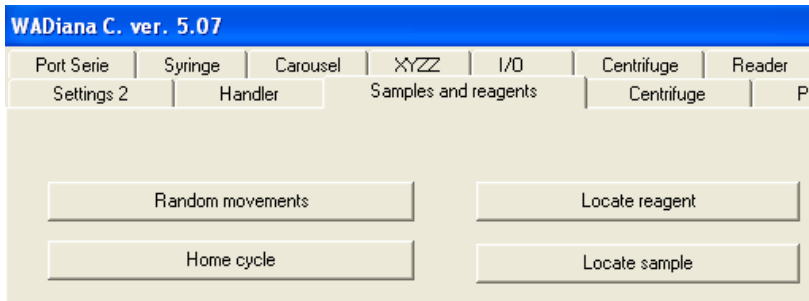
Process	Expected Results
8. Check the samples carousel window, the R1 reagent position and the reagents camera are aligned.	---
9. If the reagents carousel is slightly misaligned probably the reagent carousel offset is not properly set. Refer to procedure REGD-0000796.	---
10. If the any of the carousels are not aligned probably is due a home sensor failure. Refer to procedure REGD-0000795 for Adjusting home sensors. In case of problem persists check troubleshooting document.	---

PROCEDURE 2 - Checking settings.

Process	Expected Results
1. Access 'Settings' tab in ST software.  Picture 7	The software looks like Picture 7.
2. Check 'Reagents carousel offset' is -250 ±50. If readjustment is needed use procedure REGD-0000796.	---
3. Ensure Magnetic shaker check box is checked.	---

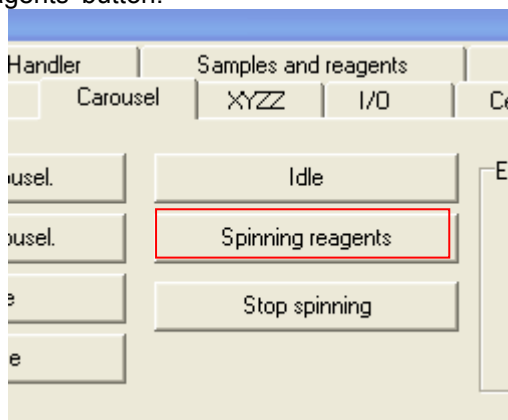
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PROCEDURE 3 - Checking positioning.

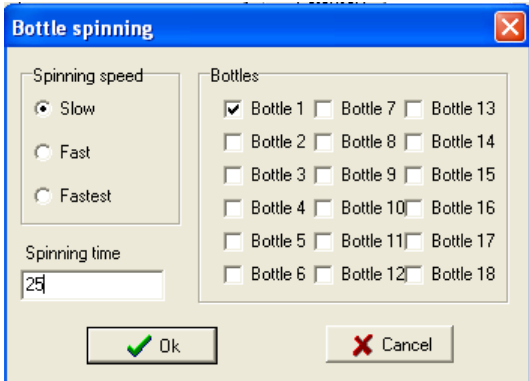
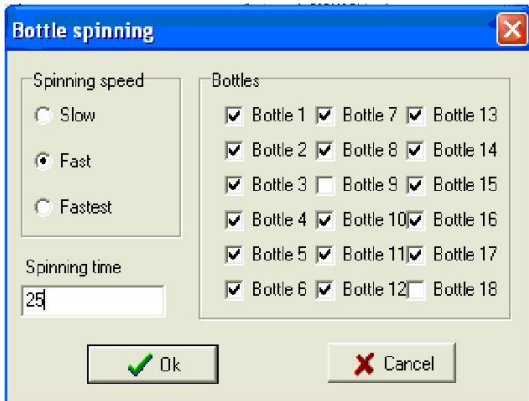
Process	Expected Results
1. Access 'Samples and reagents' tab in ST software.  Picture 8 2. Manually move both carousels to assure they aren't in home position. 3. Press 'Home cycle' button. 4. Leave the carousels moving during 5 minutes to assure the system is performing correctly. 5. Press 'Random movements' button. 6. Leave the carousels moving during 5 minutes to assure the system is performing correctly.	The software looks like Picture 8. --- Carousels spin counterclockwise up to 1 turn many times until 'stop' button is pressed. --- Carousels move randomly until 'stop' button is pressed. ---

PROCEDURE 4 - Reagents spinning.

Note: this procedure can be executed with **No Init Technical Service software** also.

Process	Expected Results
1. Place a reflective tape in a 5ml vial cap (reagent vial tap must be correctly screwed in the reagents vial). 2. Access 'Carousel' tab in ST software. 3. Press 'Spinning reagents' button.  Picture 9	--- --- ---

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Process	Expected Results
4. Select 'Slow' spinning speed, type '25' inside 'spinning time' and select bottle position '1'.  Picture 10	---
5. Place the reagents vial indicated in step 1 at reagents carousel position nr.1 and press OK. Read reagents spinning speed with a tachometer. Speed might not be uniform due to the reagents vial friction but the average speed must be the expected one. 6. Repeat steps 4 and 5 selecting spinning speed: -Fast. -Fastest.	Slow >100rpm Fast>400rpm Fastest>500rpm
7. Place a 5ml reagents vial in all the reagents carousel positions (except 9 and 18) 8. Select 'fast' speed, 25 seconds and all the bottle position (except 9 and 18):  Picture 11	---
9. Check that all reagent vials are able to spin.	All reagent vials are able to spin.
10. Repeat steps 7,8and 9 with a 10ml reagent vial.	

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/06	<i>Document REGD-0000797_v2 replaces document 57025V0M000.</i>	Document reference changed.
3.0	2011/06/11	Added previous version 'data', 'change' and 'rationale' inside document change history section.	Update change history for previous document history modification.
3.0	2011/06/11	<i>Document REGD-0000797_v3 replaces document REGD-0000797_v2.</i>	Document version changed.
3.0	2011/06/11	Deleted REGD-0000788 reference.	Document available for DG-57 and DG517.
3.0	2011/06/11	Document title.	Commercial reference substituted by project reference.
3.0	2011/06/11	Document header.	Increase information to easy document search.
4.0	See document header.	Pictures updated with last software version.	Update document with current software version.
4.0	See document header.	Procedure 4 –reagents spinning-updated.	Check correct reagents spinning speed.

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Print date:	31-oct-2012	Page	8 of 8
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